

Total 15 marks quiz

Q1. Why symmetric encryption rarely used alone on the internet? **[1 mark]**

- A) It is slow
- B) Cannot encrypt large data
- C) **Secure key distribution is difficult**
- D) Cannot provide integrity

Q2- PUCIT CMS stores student grades in a database server. Recently, BS IT students complained that **[4 marks]**

- a) Some grades were changed without permission **[Integrity violation]**
- b) At times, the system was unreachable **[Availability violation]**
- c) Hackers attempted to access grade reports **[Confidentiality violation]**

Identify which element(s) of the CIA Triad _____ are violated in each situation

Q3. A system administrator at PUCIT is troubleshooting a network issue. By mistake, Zain Ali applied an incorrect firewall configuration that blocks all incoming traffic to the company's primary web server for 30 minutes until the error is fixed. **[2 marks]**

- a) Which component(s) of the CIA Triad was affected? **[Availability violation]**
- b) Was this a security breach or a security incident? And why? **[security incident]**

Q4. An angry employee, Ruby Tahir, used her access to the customer database and slightly modified the "annual sales report" document just before it was sent to the company's investors. She changed a few numbers to make her former team's performance look worse. **[3 marks]**

- a) Which component of the CIA Triad was directly violated? **[Integrity violation]**
- b) Recommend something where a receiver can verify the file is the same as sent by the sender ? **[Hashing/fingerprinting]**
- c) To protect the report in future, Information Security officer Mr Zain Ali kept the file in his laptop only. Is there a problem with this? **[Availability violation]**
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Q5. What is a Ransomware Attack and how can you prevent it? [2 marks]

Steal data or encrypt then ask money for decryption keys. Solution can be 0-trust etc

Q6. Please switch ON your laptops and then try to compromise Krypton. Please write passwords of all levels you have hacked. For level 1, you can use following

Ssh krypton1@krypton.labs.overthewire.org -p 2231 [3 marks] rest of the question will not be marked, only check if this command is written or used.

Username: krypton1,

Host: krypton.labs.overthewire.org

Port: 2231

Password (Base64) S1JZUFRPTkITR1JFQVQ=

Level 1

Level 2

Level 3

Level 4

